



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Subject: Programming with Java

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Div: CSE A1

Batch:2025-29

ASSIGNMENT No : 10

TITLE :Write a Java program to demonstrate the use of abstract classes and abstract methods

Problem Statement

Write a Java program to demonstrate the use of abstract classes and abstract methods -----

Learning Objectives

To implement abstraction using abstract classes and abstract methods.

Theory

Abstract classes provide partial implementation of a program and cannot be instantiated directly. They may contain both abstract and concrete methods. Abstract methods declare functionality that must be implemented by subclasses. Abstraction hides implementation details while exposing only essential functionality. This improves code flexibility and maintainability in object-oriented programming.

Outcome

The program successfully created an abstract class containing an abstract method, and two subclasses provided their own concrete implementations. Objects of the subclasses correctly executed their respective implementations when the abstract method was called.

Conclusion

Abstraction through abstract classes allows a common template to be defined while leaving implementation-specific details to subclasses. This experiment demonstrated how abstract classes enforce a consistent structure while allowing flexible, varied behavior.

Exercise

1. Create an abstract **Payment** class and implement Credit Card and UPI payment classes.?

```
1  abstract class Payment {
2      |    abstract void pay(double amount);
3      |
4  }
5  class CreditCardPayment extends Payment {
6      |    void pay(double amount) {
7      |        |    System.out.println("Paid " + amount + " using Credit Card");
8      |        |
9      |    }
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```

1  abstract class FoodOrder {
2      abstract double calculateBill();
3  }
4  class DineInOrder extends FoodOrder {
5      double amount;
6      DineInOrder(double amount) { this.amount = amount; }
7      double calculateBill() {
8          return amount + (amount * 0.10); // 10% service charge
9      }
10 }
11 class TakeAwayOrder extends FoodOrder {
12     double amount;
13     TakeAwayOrder(double amount) { this.amount = amount; }
14     double calculateBill() {
15         return amount; // no service charge
16     }
17 }
18 public class FoodOrderDemo {
19     Run | Debug
20     public static void main(String[] args) {
21         FoodOrder dineIn = new DineInOrder(amount: 500);
22         FoodOrder takeAway = new TakeAwayOrder(amount: 500);
23         System.out.println("Dine-In Bill: " + dineIn.calculateBill());
24         System.out.println("Take-Away Bill: " + takeAway.calculateBill());
25     }
26 }

```

```

Dine-In Bill: 550.0
Take-Away Bill: 500.0

```
